

Name: Himanshu Sharma (230474)

PHY 412: Assignment 2

Ans: $H(\{q_i, p_i\}) = \sum_{i=1}^N \left[\frac{p_i^2}{2m} + \frac{m\omega^2 q_i^2}{2} \right]$

(a) canonical transforms:

$$q_i' \equiv \sqrt{m\omega} q_i$$

$$p_i' = \frac{p_i}{\sqrt{m\omega}}$$

$$\Rightarrow E = H(\{q_i', p_i'\}) = \sum_{i=1}^N \frac{\omega p_i'^2}{2} + \frac{\omega q_i'^2}{2}$$

$$\Rightarrow E = \frac{\omega}{2} \sum_{i=1}^N (p_i'^2 + q_i'^2) \Rightarrow \text{sphere in } 2N\text{-dim}$$

$$\Rightarrow R = \sqrt{2E/\omega}$$

vol of hypersphere: $\frac{\pi^{d/2} R^d}{(d/2)!} = \frac{\pi^N (2E/\omega)^N}{N!}$

$$\Rightarrow \Omega \approx \left(\frac{2\pi E}{h\omega} \right)^N \frac{1}{N!}$$

$$\Rightarrow S = K_B \ln \Omega = K_B \left[N \ln \left(\frac{2\pi E}{h\omega} \right) - \ln N! \right]$$

$$\ln N! \approx N \ln N \Rightarrow S = K_B N \left(\ln \left(\frac{2\pi E}{h\omega} \right) - \ln N + 1 \right)$$

$$\Rightarrow S = K_B N \ln \left(\frac{2\pi E e}{h\omega N} \right)$$

$$(b) \frac{1}{T} = \left. \frac{\partial S}{\partial E} \right|_N = \frac{K_B N (h\omega N)}{2\pi E e} \cdot \frac{2\pi e}{h\omega N} = \frac{N K_B}{E}$$

$$\Rightarrow T = E / N K_B \Rightarrow \boxed{E = N K_B T}$$

$$\boxed{C = N K_B}$$

$$\begin{aligned}
 (c) \quad P(p_1, q_1) &= \frac{\Omega(N-1, E_{N-1})}{\Omega(N, E)} \times \frac{1}{h} \\
 &= N! \left(\frac{h\omega}{2\pi E} \right)^N \frac{1}{h} \left(\frac{2\pi}{h\omega} \right)^{N-1} \frac{1}{(N-1)!} \left[E - \frac{p_1^2}{2m} - \frac{m\omega^2 q_1^2}{2} \right]^{N-1} \\
 &= \frac{N\omega}{2\pi} \frac{1}{E^N} \left[E - \frac{p_1^2}{2m} - \frac{m\omega^2 q_1^2}{2} \right]^{N-1} \\
 &= \frac{N\omega}{2\pi E} \left[1 - \frac{p_1^2}{2mE} - \frac{m\omega^2 q_1^2}{2E} \right]^{N-1}
 \end{aligned}$$

$$E = NK_B T$$

$$\Rightarrow P(p_1, q_1) \approx \frac{\omega}{2\pi K_B T} \left[1 - \frac{p_1^2}{2mNK_B T} - \frac{m\omega^2 q_1^2}{2NK_B T} \right]^N$$

$$(1 + \epsilon)^N \approx 1 + N\epsilon \approx \exp(N\epsilon)$$

$$\Rightarrow \boxed{P(p_1, q_1) \approx \frac{\omega}{2\pi K_B T} \exp \left\{ -\frac{p_1^2}{2mK_B T} - \frac{m\omega^2 q_1^2}{2K_B T} \right\}}$$

$$\left\langle \frac{p^2}{2m} \right\rangle = \int P(p, q) \frac{p^2}{2m} dp dq = \frac{K_B T}{2}$$

$$\left\langle \frac{m\omega^2 q^2}{2} \right\rangle = \int P(p, q) \frac{m\omega^2 q^2}{2} dp dq = \frac{K_B T}{2}$$

Ans 2 $\Rightarrow \Omega = \frac{1}{N! h^{3N}} \int d^{3N} p \int d^{3N} q$

$$H = \sum_{i=1}^{3N} \frac{1}{2m} p_i^2 \Rightarrow R = \sqrt{2mE}$$

$$\text{volume} = \frac{\pi^{3N/2} R^{3N}}{(3N/2)!} = \frac{(2mE\pi)^{3N/2}}{(3N/2)!}$$

$$\int d^{3N} q = V(V-\omega)(V-2\omega) \dots (V-(N-1)\omega)$$

$$(V-a\omega)(V-(N-a)\omega) \approx (V-N\omega/2)^2$$

$$\Rightarrow \int d^{3N} q = (V-N\omega/2)^N$$

$$\Rightarrow \Omega = \frac{(2mE\pi)^{3N/2}}{(3N/2)!} (V-N\omega/2)^N \frac{1}{N!}$$

$$\Rightarrow S = K_B \ln \Omega = K_B \left[N \ln (V-N\omega/2) + \frac{3N}{2} \ln (2mE\pi) - 3N/2 \ln (3N/2) + 3N/2 - \frac{N \ln N}{+N} \right]$$

$$\Rightarrow S = NK_B \ln \left(\frac{(V-N\omega/2) (2mE\pi)^{3/2} e^{3/2}}{(3N/2)^{3/2} N} \right)$$

$$\Rightarrow S = NK_B \ln \left[\left(\frac{4mE\pi}{3N} \right)^{3/2} \left(V - \frac{N\omega}{2} \right) \frac{e}{N} \right]$$

$$(b) \quad \frac{P}{T} = \left. \frac{\partial S}{\partial V} \right|_{N,E} = \frac{NK_B}{V - \frac{N\omega}{2}}$$

$$\Rightarrow \boxed{P(V - N\omega/2) = NK_B T}$$

$$(c) \quad K_T = -\frac{1}{V} \left. \frac{\partial V}{\partial P} \right|_T, \quad V - \frac{N\omega}{2} = \frac{NK_B T}{P}$$

$$\Rightarrow \left. \frac{\partial V}{\partial P} \right|_T = -\frac{NK_B T}{P^2}$$

$$\Rightarrow K_T = -\frac{1}{V} \left(-\frac{NK_B T}{P^2} \right) = \frac{NK_B T}{P^2 V} > 0$$

$$\text{as } T, N, P, V > 0$$

Ans 3 →

(a) The ground state energy of $E = E_{\min} = 0$ is obtained for 2^N configurations. largest microstate energy is, NE is unique.

(b) $N_z = E/\epsilon$

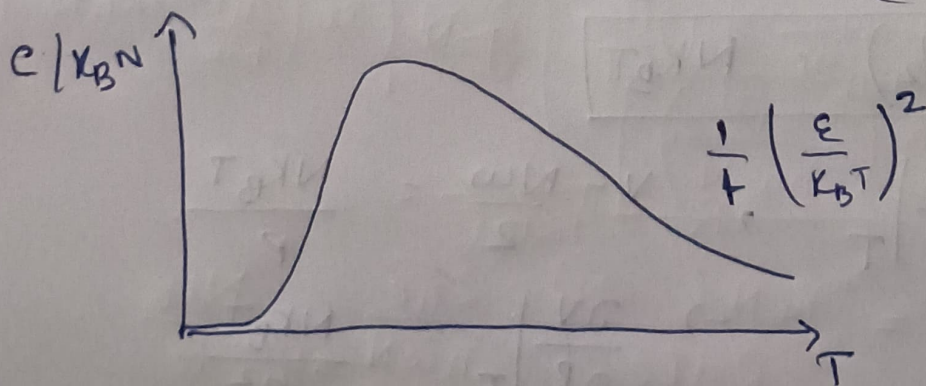
$$\Omega(E, N) = \frac{N!}{N_z! (N - N_z)!} \frac{1}{2^{N - N_z}}$$

$$S(E, N) = -Nk_B \left(\frac{E}{NE} \ln \frac{E}{NE} + \left(1 - \frac{E}{NE} \right) \ln \left(1 - \frac{E}{NE} \right) \right) + k_B (N - E/\epsilon) \ln 2$$

(c) $T = \left(\partial S / \partial E \right)_N^{-1} = \left(-\frac{k_B}{E} \ln \left(\frac{E}{NE - E} \right) - \frac{k_B \ln 2}{E} \right)^{-1}$

$$\Rightarrow E = \frac{NE}{2 \exp \left(\frac{E}{k_B T} \right) + 1}$$

$$c = \frac{dE}{dT} = Nk_B \left(\frac{E}{k_B T} \right)^2 \frac{2 \exp \left(E/k_B T \right)}{\left(1 + 2 \exp \left(E/k_B T \right) \right)^2}$$



(d) $P(\vec{r}_1 = z) = \Omega(E - \epsilon, N - 1) / \Omega(E, N)$
 $= \left[2 \exp \left(E/k_B T \right) + 1 \right]^{-1}$

(e) $E_{\max} = NE/3$

Ans 4 -

$$(a) P_i \propto e^{-\beta E_i}, \quad \beta = 1/k_B T$$

$$P_1/P_2 = \exp\{\beta(E_2 - E_1)\}$$

$\rightarrow$ As $T \rightarrow \infty : e^0 = 1 \Rightarrow P_1 = P_2 = 1/2$ (maximal entropy)

$\rightarrow$ As $T \rightarrow 0$: Since $E_2 > E_1$, $\exp\{\beta(E_2 - E_1)\} \rightarrow \infty$.

$\Rightarrow P_1 \gg P_2$ and system settles to ground state

$$(b) P_1 = \frac{\exp(-\beta E_1)}{e^{-\beta E_1} + e^{-\beta E_2}}, \quad P_2 = \frac{\exp(-\beta E_2)}{e^{-\beta E_1} + e^{-\beta E_2}}$$

$$\langle E \rangle = \sum P_i E_i = \frac{E_1 \exp(-\beta E_1) + E_2 \exp(-\beta E_2)}{\exp(-\beta E_1) + \exp(-\beta E_2)}$$

Ans 5 -

$$(a) Z_1 = \sum \exp(-\beta E_i) = e^{-\beta(0)} + e^{-\beta E} = 1 + e^{-\beta E}$$

$$Z(T, N) = (1 + e^{-\beta E})^N$$

$$(b) \text{ long (2a), } P_L = 1/(1 + e^{-\beta E})$$

$$\text{short (a), } P_S = e^{-\beta E} / (1 + e^{-\beta E})$$

$$(c) L = \sum_{i=1}^N l_i, \quad \langle L \rangle = N \langle l \rangle$$

$$\langle l \rangle = P_L(2a) + P_S(a) = \frac{2a + a e^{-\beta E}}{1 + e^{-\beta E}}$$

$$\langle L(T, N) \rangle = Na \left(\frac{2 + e^{-\beta E}}{1 + e^{-\beta E}} \right)$$

$$(d) \text{Var}(l) = \langle l^2 \rangle - \langle l \rangle^2$$

$$\langle l^2 \rangle = P_L(2a)^2 + P_S(a)^2 = \frac{4a^2 + a^2 e^{-\beta E}}{1 + e^{-\beta E}}$$

$$\Rightarrow \text{Var}(l) = \frac{a^2 e^{-\beta E}}{(1 + e^{-\beta E})^2}$$

$$\langle L(T, N)^2 \rangle_c = \frac{Na^2 \exp(-\beta \epsilon)}{(1 + \exp(-\beta \epsilon))^2}$$

Ans 6

(A) Canonical Partition Function

No. of ways to chose the sites $N_0 C_N$

$$Z(T, N, N_0) = N_0 C_N [a(T)]^N$$

$$F = -k_B T \ln Z = -k_B T [\ln N_0! - \ln N! - \ln (N_0 - N)! + N \ln a(T)]$$

$$\Rightarrow F \approx -k_B T (N_0 \ln N_0 - N \ln N - (N_0 - N) \ln (N_0 - N) + N \ln a(T))$$

$$\mu = \left(\frac{\partial F}{\partial N} \right)_{T, N_0} = -k_B T [-\ln N - 1 + \ln (N_0 - N) + 1 - \ln a(T)]$$

$$\boxed{\mu = k_B T \ln \left(\frac{N}{(N_0 - N) a(T)} \right)}$$

(B) Grand Partition Function

Single-site partition function: $\xi = 1 + a(T) e^{\beta \mu}$

Total grand partition function: $Z = \xi^{N_0} = (1 + a(T) e^{\beta \mu})^{N_0}$

Average occupancy, $N = k_B T \frac{\partial \ln Z}{\partial \mu}$

$$= N_0 \frac{a(T) e^{\beta \mu}}{1 + a(T) e^{\beta \mu}}$$

$$\Rightarrow e^{\beta \mu} = \frac{N}{(N_0 - N) a(T)} \Rightarrow \boxed{\mu = k_B T \ln \left(\frac{N}{(N_0 - N) a(T)} \right)}$$

Ans 7

Grand Partition Function:

$$\mathcal{Z}(T, \mu, V) = \sum_{N=0}^{\infty} \frac{1}{N!} (Z_1 e^{\beta\mu})^N = \exp(Z_1 e^{\beta\mu})$$

For monatomic ideal gas, $Z_1 = V/\lambda^3$

$$\text{where } \lambda = \sqrt{\frac{h^2}{2\pi m k_B T}}$$

$$\text{Grand Potential } \Phi = -k_B T \ln \mathcal{Z} = -k_B T Z_1 e^{\beta\mu}$$

$$\text{using } \Phi = -PV : P = -\frac{\Phi}{V} = \frac{k_B T Z_1 \exp(\beta\mu)}{V}$$

$$\text{average particle number } N: N = -\left(\frac{\partial \Phi}{\partial \mu}\right)_{T,V} = Z_1 e^{\beta\mu}$$

$$e^{\beta\mu} = \frac{N}{Z_1} \Rightarrow \mu = k_B T \ln\left(\frac{N}{Z_1}\right) = k_B T \ln\left(\frac{N \lambda^3}{V}\right)$$